

pole mount; and advanced algorithms for event and system analysis.

Performance optimisation at source

It is interesting to note that the role of technology begins at the very beginning of the value chain. Performance optimisation measures taken at the fuel exploration and energy production sites also play a major role in improving the energy availability and costs.

"Increasingly, oil and gas companies are investing in technologies to realise the potential of the 'digital oilfield.' Advanced analytics, Big Data, complex event processing and high-performance computing are being used to enable organisations to optimise performance in oil drilling and production. Technology companies like ours are investing in solutions designed for exploration and production companies in the energy sector that reduce exploration risk, cost per barrel and improve performance of oilfield assets," says Sanjay Kalra, vice president and head of delivery-Energy and Utilities, iG-ATE, the integrated Technology and Operations (iTOPS) company providing business outcome-based solutions.

Tools for developing innovative solutions

As with any emerging field, the energy sector also needs efficient tools to design, develop, test and deploy new technologies.

"The key to building an innovative solution is the ability to quickly prototype and prove its utility, adhering to the standards and protocols that are followed in the energy sector, and meeting the reliability/durability requirements," says Mohanram.

"This brings us to the discussion about tools that can help developers build these systems. There is a significant leverage of cutting-edge PC technologies in today's energy management systems. This trend has channelled people towards commercial off-the-shelf (COTS) hardware platforms and high-level software tools that provide enough abstraction

for a domain expert to put together the system. The COTS hardware platform takes care of adhering to the mechanical and electrical standards and the high-level software provides readymade libraries for the protocols followed in the Industry."

Reconfigurable embedded instrumentation and control systems, such as National Instruments' CompactRIO, provide an ideal combination of technologies and features to address the most difficult smart grid challenges. Powered by NI LabVIEW and reconfigurable field-programmable gate arrays (FPGAs), these user-programmable, field-updatable smart devices can perform multiple digital signal processing and control tasks in parallel and in real-time. Furthermore, modern analogue-to-digital converters (ADCs) and sensors provide high-fidelity electrical measurements while synchronising on a global scale. In addition, emerging network communication protocols such as IEC 61850 are being defined to ensure network interoperability and compatibility from the smart sensor to the cloud.

Double-click on energy

Slowly, we can see that technology is changing things in the energy sector. Dr Kumar points out that today's companies are taking an active interest in making their operations more efficient and effective. Individuals are becoming more and more aware of their power usage and ways to conserve it. So far, energy production had been centralised to adapt to the demand variation. Renewable energy generation is enabling an era of distributed generation that can integrate with the grid, which can help in reducing transmission and distribution losses as well as cater to peak demand. Attempts are being made to make the transmission and distribution systems more intelligent and flexible. HV/MV and MV/LV substations with indoor solutions for limited space, urban and outdoor areas, as well as unmanned solutions for rural and remote landscapes guaranteeing safety of people and assets,

are an illustration of this. Substation automation and renewable energy connectivity are other examples.

"However, we feel India still has a long way to go when we talk about smart monitoring solutions. Technology is available here, but a wide acceptance from common people is lacking. The concept of energy monitoring and management is growing fast in the country but a large chunk of buildings and industry still lack these latest technologies," says Dr Kumar.

It is obvious that all the required technology is here. What we need is a change in mind-set. As Prof. Bhavne innovatively terms it, we need to double-click on energy!

"Ambient energy – sunlight – is everywhere and plentiful, yet when we speak about energy we typically mean fuels – coal, oil, gas, wood, or nuclear – and often their use for generating electricity or for transport. Instead, we could be talking about use – cooling or refrigeration, heating water and air, and mobility (public and personal transport) – and less about litres and kilowatt-hours. Can we parse the electricity bill into water heating, cooking, air-conditioning and transport? Why do we measure kilowatt-hours in undifferentiated bulk each month? The point is, energy – as we use the term – is amorphous and not enlightening; we need to double-click on it," says Prof. Bhavne.

"We use a lot of appliances today – refrigerators, microwaves, washing machines, lighting, air-conditioners. We don't have month-to-date energy use and its cost. We don't know easily how much each hot-water shower costs. Or the hourly cost of refrigeration. If we can have per second mobile phone billing, surely this can be done for electricity? It is interesting to speculate why this is not the case. The technical solution is not difficult. It has a big role to play. I think this area is ripe for technical and business model innovation. The risk is timing," he adds. Food for your thought! ●

The author is a technically-qualified freelance writer, editor and hands-on mom based in Chennai